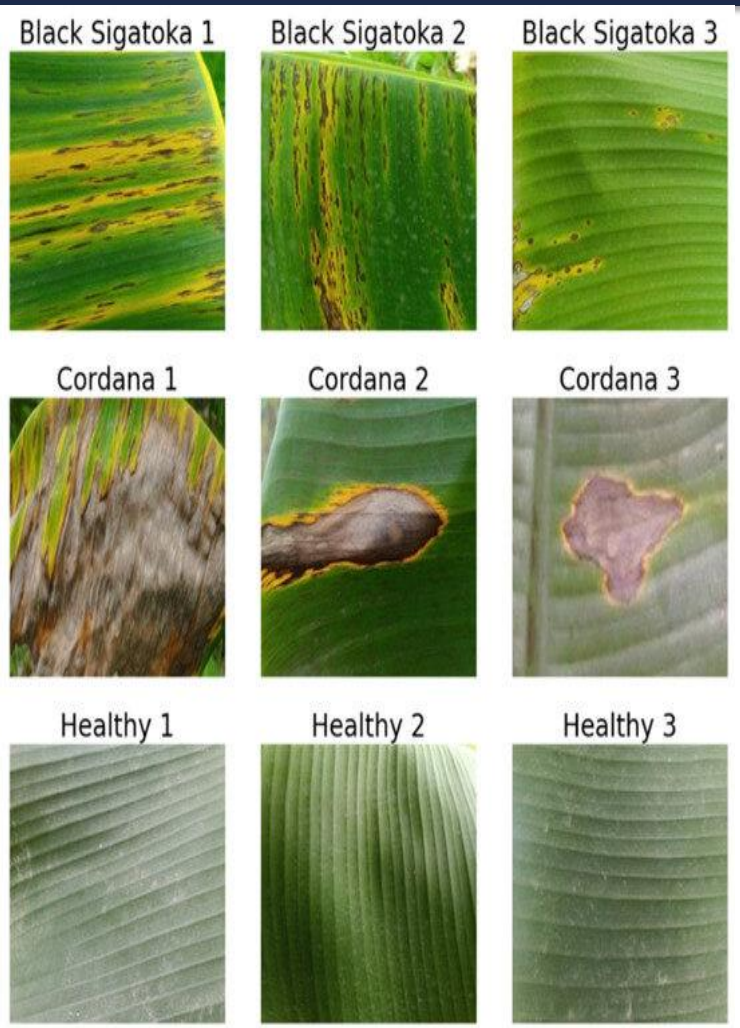
CS-AIML 231: Introduction to AI & Machine Learning

University of Somalia (UNISO) · Faculty of Computer Science & IT

"AI is not magic. It is math, data, and code."

Instructor: Eng. **Abdulkadir Dahir** | Semester 1, Week 1

Story: A Farmer in Afgooye



The Problem:

"Last year, a farmer in Afgooye region lost 40% of his banana crop. The disease was treatable — but he didn't detect it early enough. By the time he saw the spots on the leaves, it was too late."

The AI Solution:

"Today, a simple AI model on a phone can identify banana leaf disease from a photo. The farmer spots it in time, treats the crop, saves his income."

This is not science fiction. This is what you will learn to build.

What is Artificial Intelligence?

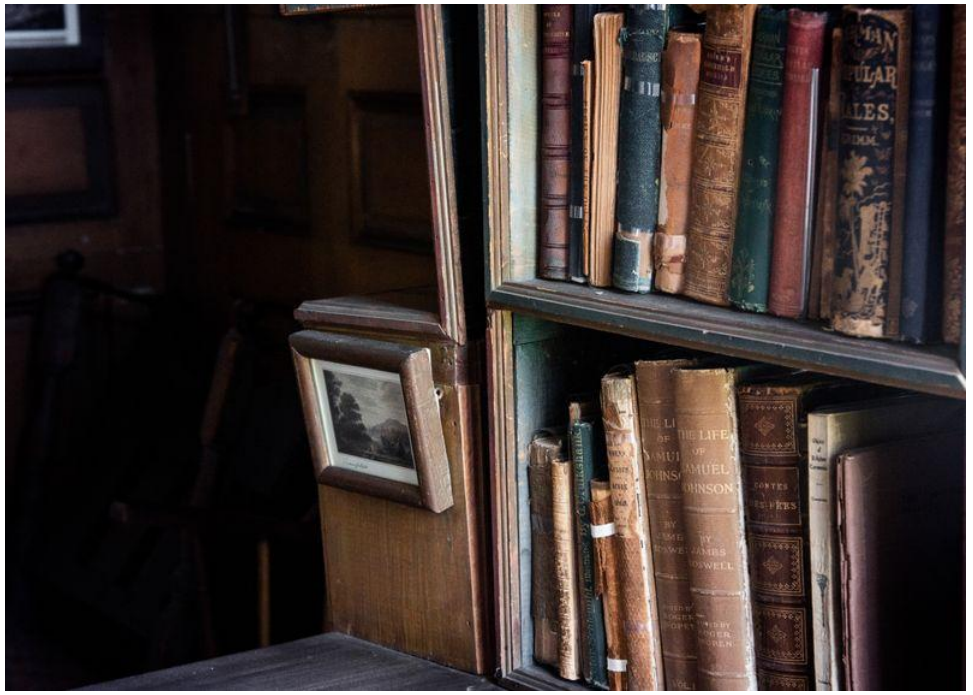
Definition:

AI is any system that performs tasks requiring human intelligence — seeing, hearing, speaking, deciding, learning.

Analogy:

"AI is like a child learning to read. At first it knows nothing. You show it examples (letters, words), and slowly it recognizes patterns. Eventually it reads on its own."

The child does NOT have rules memorized. It learns from EXAMPLES.



Somali context: Mobile money fraud detection, voice assistants, Google Translate

Three Types of AI

Narrow AI

Good at ONE thing only.
Like a calculator — perfect at math,
but cannot write a poem.

**Calculator, chess engine,
voice assistants, ChatGPT**

General AI

Could do ANYTHING a human can.
Does not exist yet.
Like a person who can be a doctor,
engineer, artist, teacher — all in one.

**This is the GOAL.
We are not here yet.**

Super AI

Smarter than ALL humans.
Purely speculative.
Like a god-like mind.

**Science fiction for now.
Focus on what exists: Narrow AI.**

ChatGPT FEELS general, but it is narrow AI. It can write, but it cannot drive a car or feel emotions.

AI vs ML vs Deep Learning

Think of it like a set of nesting dolls.



Key Insight:

Most of what you see in the news (ChatGPT, image generators) is deep learning. But classical ML (regression, decision trees) is still widely used and often more appropriate for small datasets.

AI = any machine intelligence

ML = AI that learns from data

DL = ML with multi-layer neural network



History of AI — A Story of Hope and Disappointment

1950	**Turing Test**	"Can machines think?" — Alan Turing
1956	**AI Born**	Dartmouth Conference — AI named as a field
1980s–90s	**AI Winters**	Funding dried up. Overpromising, underdelivering.
2012	**Deep Learning**	AlexNet wins ImageNet — neural networks return
2016	**AlphaGo**	AI beats Go world champion — worldwide shock
2020s	**Generative AI**	ChatGPT, GPT-4, Gemini — AI for everyone

Two Ways to Solve a Problem

Recipe Book (Traditional Programming)

Rules + Data → Answers

```
if temperature > 30°C:
    wear_light_clothes()
else:
    wear_jacket()
```

You write EVERY rule. The machine follows exactly.

Gardening (Machine Learning)

Answers + Data → Rules

[Show 100 days of weather + what people wore]

Machine: "I see a pattern!
When temp > 30°C, people wear light clothes."

The machine DISCOVERS the rules.

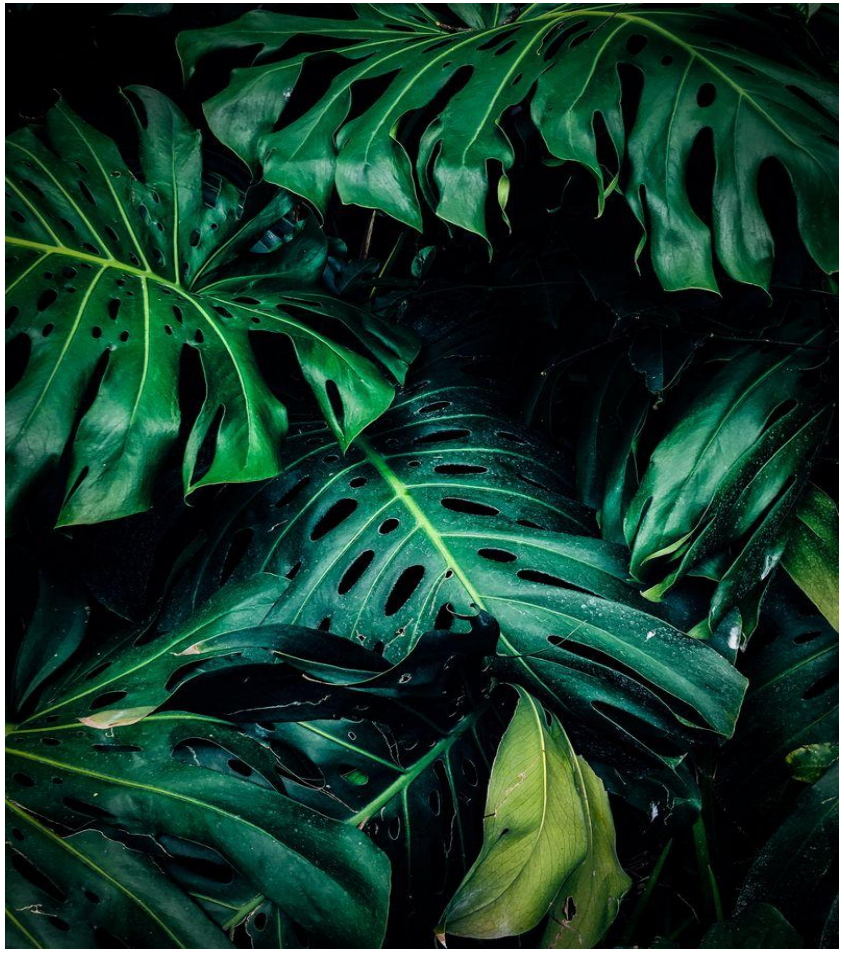
Cooking analogy: A recipe tells you exactly what to do. Learning to cook by tasting — you adjust, experiment, improve.

What is Machine Learning?

Analogy: Growing a Plant

"Traditional programming is like a recipe book. You write exact steps: 'Add 2 cups flour, bake 30 minutes.' The output is always the same.

Machine learning is like growing a plant. You don't build the plant cell by cell. You provide water, soil, sunlight — and the plant learns to grow itself."



Key Insight:

You give the machine DATA and ANSWERS.
It figures out the RULES.

Three Types of Machine Learning



Analogy: Navigating to Mogadishu

"How do you get to a destination? Three ways."

Supervised

Walking with a map
You know the destination.
Every step is labeled: 'turn left,' 'go straight.'
Example: Predict crop yield from rainfall.
You have the labels (past yields).

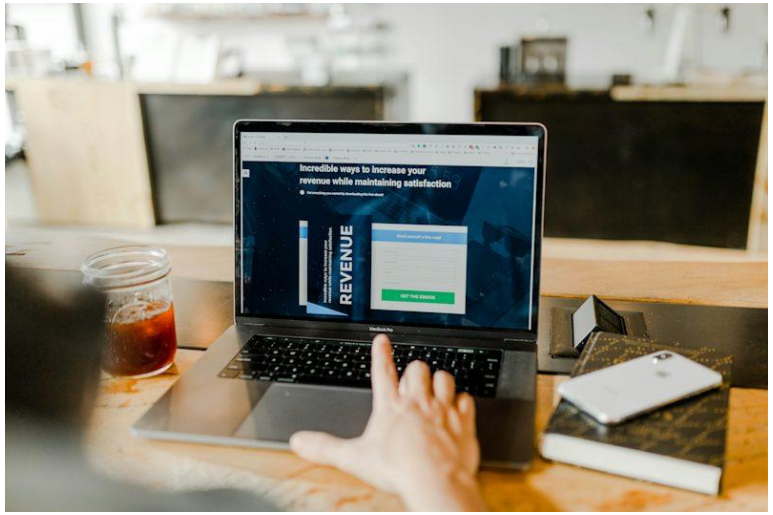
Unsupervised

Exploring without a map
You find hidden structure.
You group things by similarity.
Example: Group customers by behavior.
You don't know the groups beforehand.

Reinforcement

Learning by trial and error
Like learning to ride a bike.
You fall, adjust, try again.
Rewards guide you.
Example: A drone learning to fly.

AI in Somalia — Real Problems, Real Solutions



This is NOT tokenism. Somalia has real AI applications:

AGRICULTURE: Afgooye region crop disease prediction
HEALTHCARE: Disease outbreak prediction, diagnostic aids
FINANCE: Fraud detection for Zaad, Dahabshiil
LIVESTOCK: Herding optimization, disease tracking
LANGUAGE: Somali NLP — a low-resource language challenge

You will build models that address REAL Somali problems.

Course Structure

First Half: Lecture

Theory and derivations

Whiteboard work

Algorithm walkthroughs

Discussion and Q&A

Somali context examples

Second Half : Lab

Hands-on implementation

Code from scratch

Experiments

Submission + debrief

Two-semester arc: Semester 1 = Foundations & Classical ML · Semester 2 = Deep Learning & Applications

AI Usage Policy

Analogy: Using a calculator in math class

"When you were young, you learned multiplication by hand. Then you got a calculator. Did that make you worse at math? No — it let you solve harder problems. But you still need to understand WHY multiplication works."

AI is the same. Use it to solve harder problems. But understand WHAT it does.

ENCOURAGED

- Explain concepts
- Debugging help
- Writing docs
- Test data generation

RESTRICTED

- Writing your entire lab code
- During closed-book exams
- Without disclosure
- Plagiarism of AI output

RULE: Always disclose AI assistance. You must be able to explain your code.

Assessment & Grading

Component	Weight	When
Lab Assignments (9)	50%	Weekly
Midterm Exam	25%	Week 7
Quizzes (5)	15%	Weeks 3,5,7,9,11
Participation	5%	Ongoing
Final Project	25%	Week 14
Final Exam	25%	Week 14

Grading Scale

A: 90–100 Excellent

B+: 80–89 Very Good

B: 70–79 Good

C+: 60–69 Satisfactory

C: 50–59 Pass

D: 40–49 Fail

Tools & Setup (All Free)

Python 3.10+

NumPy, Pandas, Matplotlib

scikit-learn, PyTorch

Jupyter / Google Colab (GPU!)

Git & GitHub

ChatGPT / Gemini (responsible use)

Before Second lecture:

1. Install Python 3.10+
2. pip install numpy pandas matplotlib scikit-learn jupyter
3. Create GitHub account
4. (Optional) Set up Google Colab
5. Join WhatsApp group

If you have limited hardware, use Google Colab as your primary environment.

What I Expect From You

1. Attend every session. Missing 3+ = withdrawal.
2. Arrive on time. 5 min late = absent.
3. Code every day. Even 30 minutes.
4. Understand before you code.
5. Use AI responsibly. Always disclose.
6. Ask questions. No question is stupid.
7. Back up your work. GitHub, not just your laptop.
8. Form a study group. 3–4 people, meet weekly.
9. Sleep. All-night coding produces buggy code.
10. Be honest. Model your integrity for others.

Let's Get Started

First assignment: Python refresher (ungraded, but essential)

Join WhatsApp · Install Python · Create GitHub account

Next Sunday: Mathematical Foundations for ML

Questions? Ask now or via WhatsApp anytime this week.